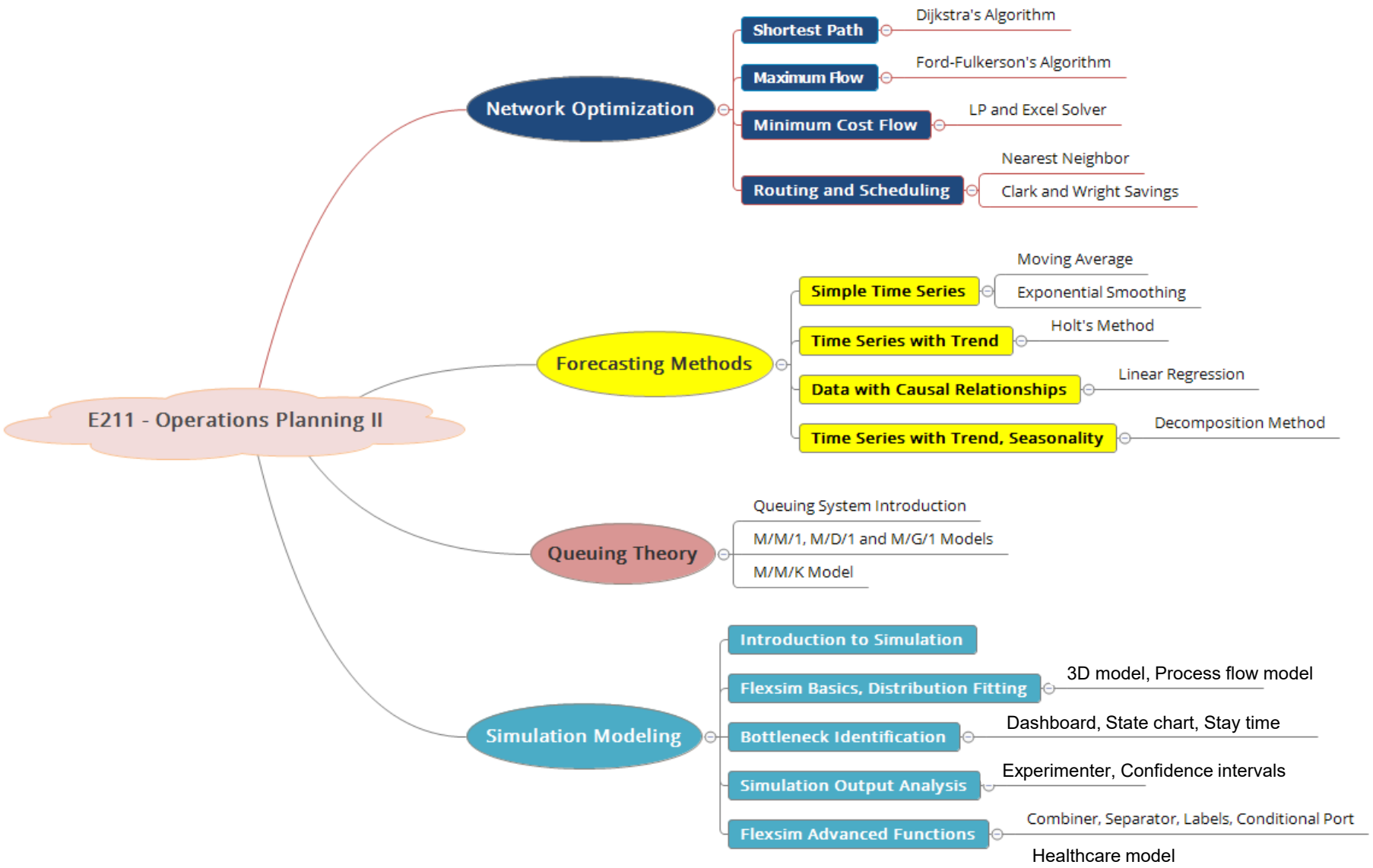
REPUBLIC POLYTECHNIC

Lesson 6
Building Your 1st FlexSim Model

E211 – Operations Planning II

REPUBLIC POLYTECHNIC

E211 Topic Tree



Recap on 3D Simulation

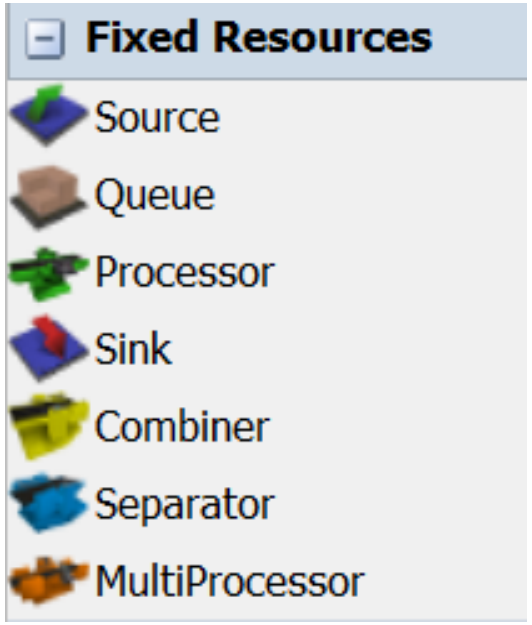
Why 3D Simulation Model

- **3D Visualization of Layout and Flow**
  A 3D model provides a realistic view of the facility, making it easier to see layout, material movement, and potential bottlenecks.

- **Accurate Material and Equipment Flow Simulation**
  The model replicates material and equipment movement in real time. This allows deep diving into the process, potentially helping to optimize workflows.

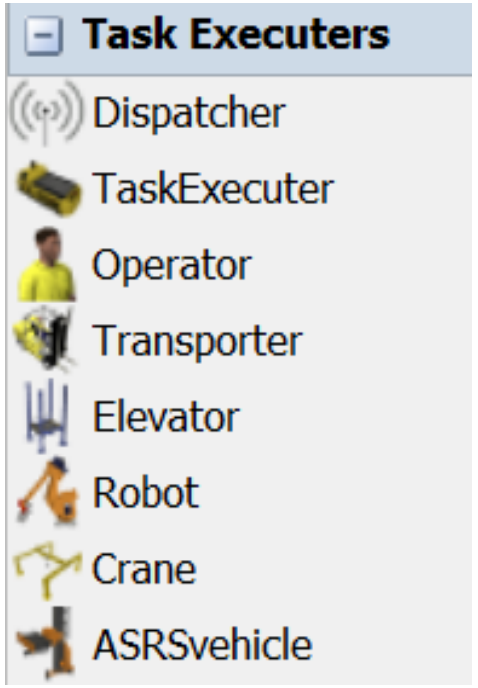
- **Space Requirement Estimation**
  Simulation of processes enables estimation of storage and activity space needs, ensuring efficient layout and minimizing wasted or overcrowded space.

Fixed Resources



- Which object creates entities?
  - **Source**
- Which object destroys entities?
  - **Sink**
- Which object processes or transforms entities?
  - **Processor, MultiProcessor, Combiner, Separator**
- Which object holds entities?
  - **Queue**

Task Executers



- **Task Executers** carry and transport flow items. They may also execute a process (e.g. an operator sets up or runs a process).

- Which object can be used to control other task executers?
  - ➢ **Dispatcher**

Conveyor and Rack

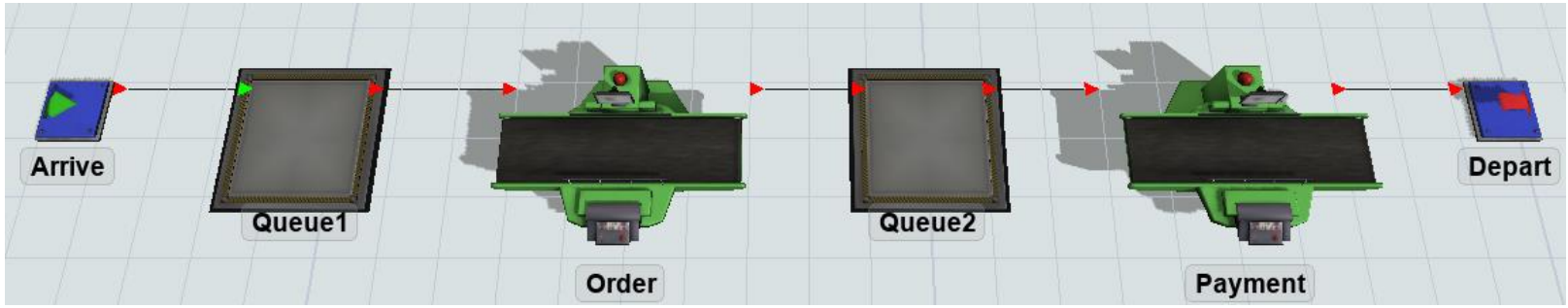
- **Conveyor** – Transport flow items
- **Rack** – Storage for flow items

Tracking Time in System (Tracked Variables)

Your 1st FlexSim Model

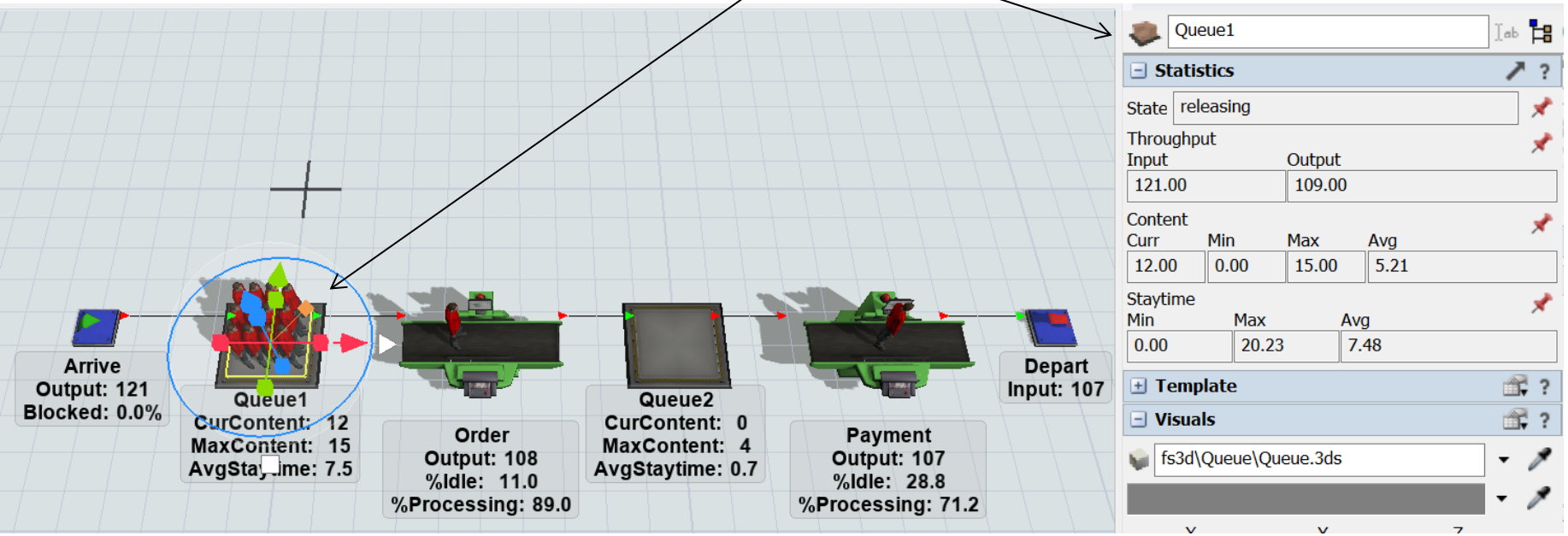
At a popular food stall, customers go through a two-stage queuing process.

Stage 1: Customer order food - served by staff #1
Stage 2: Payment - served by staff #2



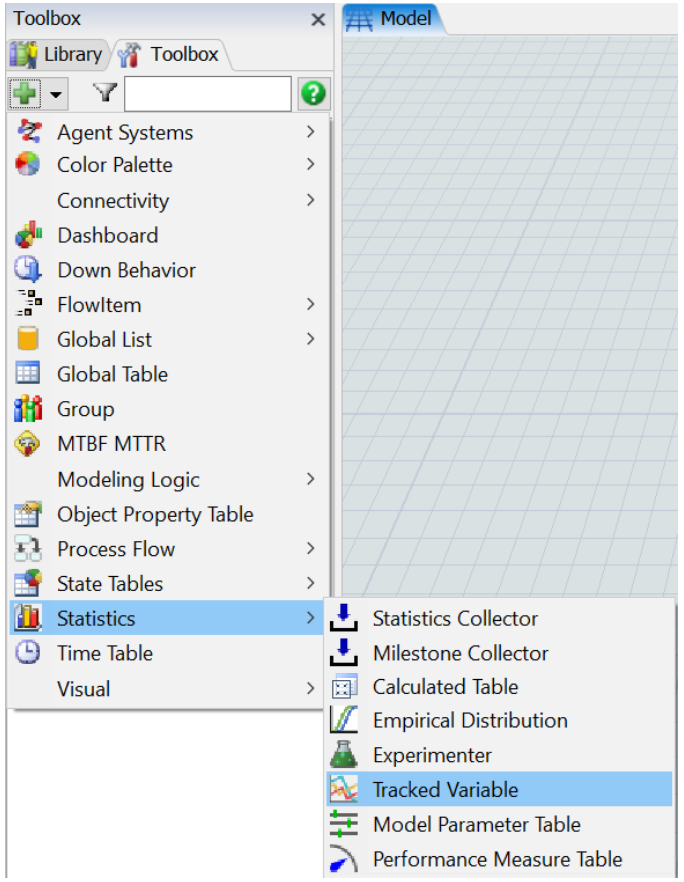
View Stats in Quick Properties Window

Click on object to view its simulation statistics in Properties window

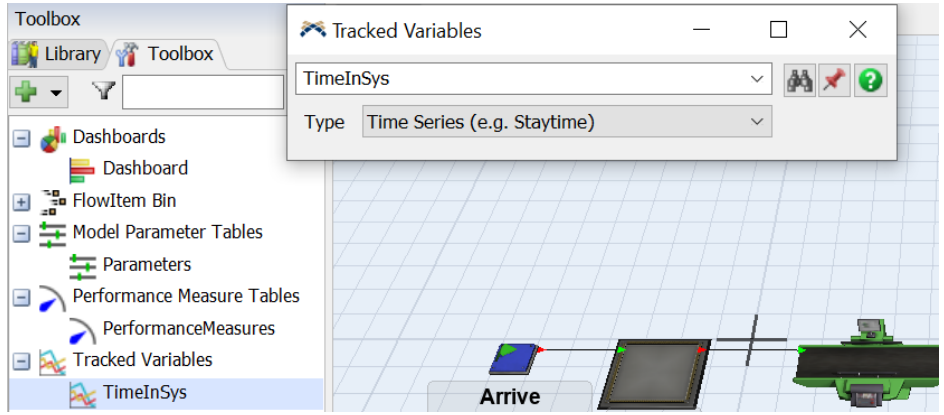


Creating a Tracked Variable

1. Go to Toolbox. Click on '+' and select Statistics → Tracked Variable



2. Rename as 'TimeInSys'

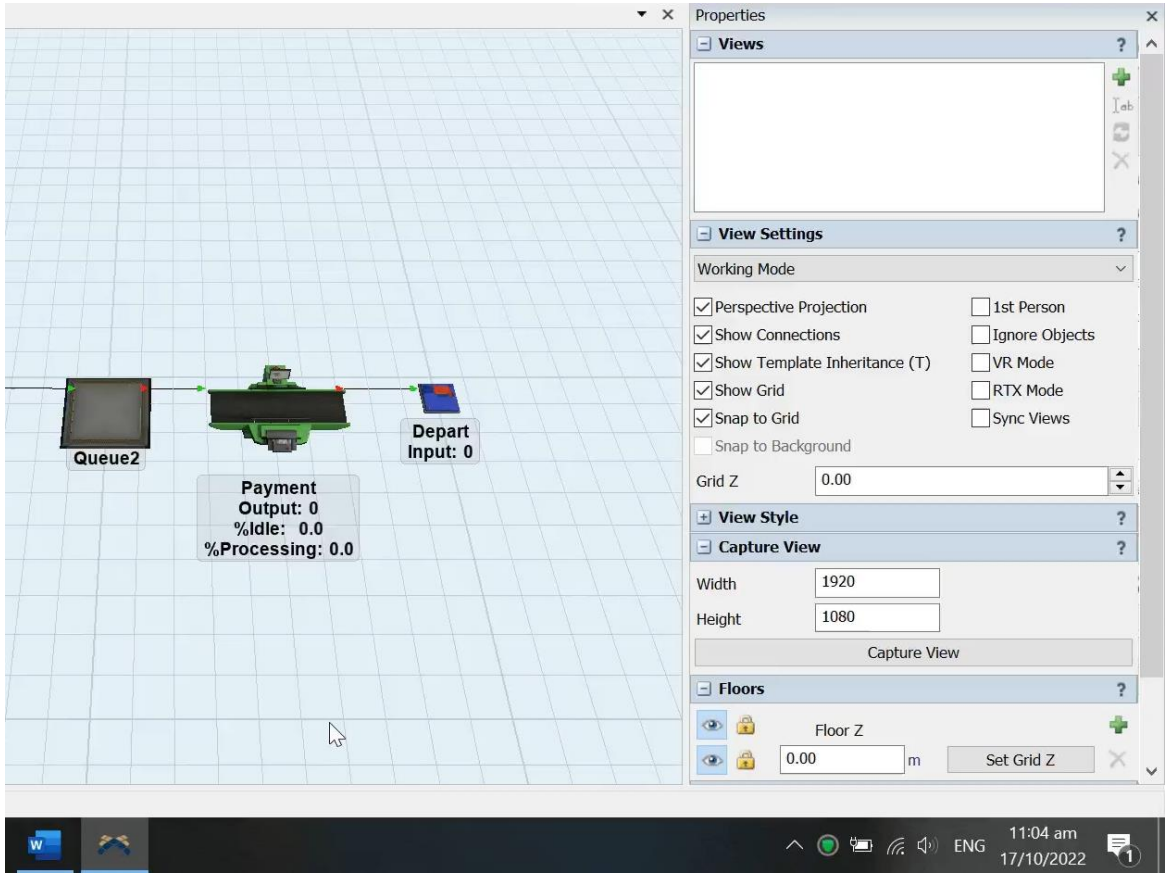


Create Time in System Variable

3. Go to Depart (Sink) properties → Trigger
Click on '+' → Click On Entry Trigger
Click on '+' → Data → Set Tracked Variable

4. Set → Select Tracked Variables → TimeInSys

5. Click option button next to 'To' → Item's Time in System



A flow item's Time in System will be tracked under the Tracked Variable (TimeInSys) when the item enters Depart sink.

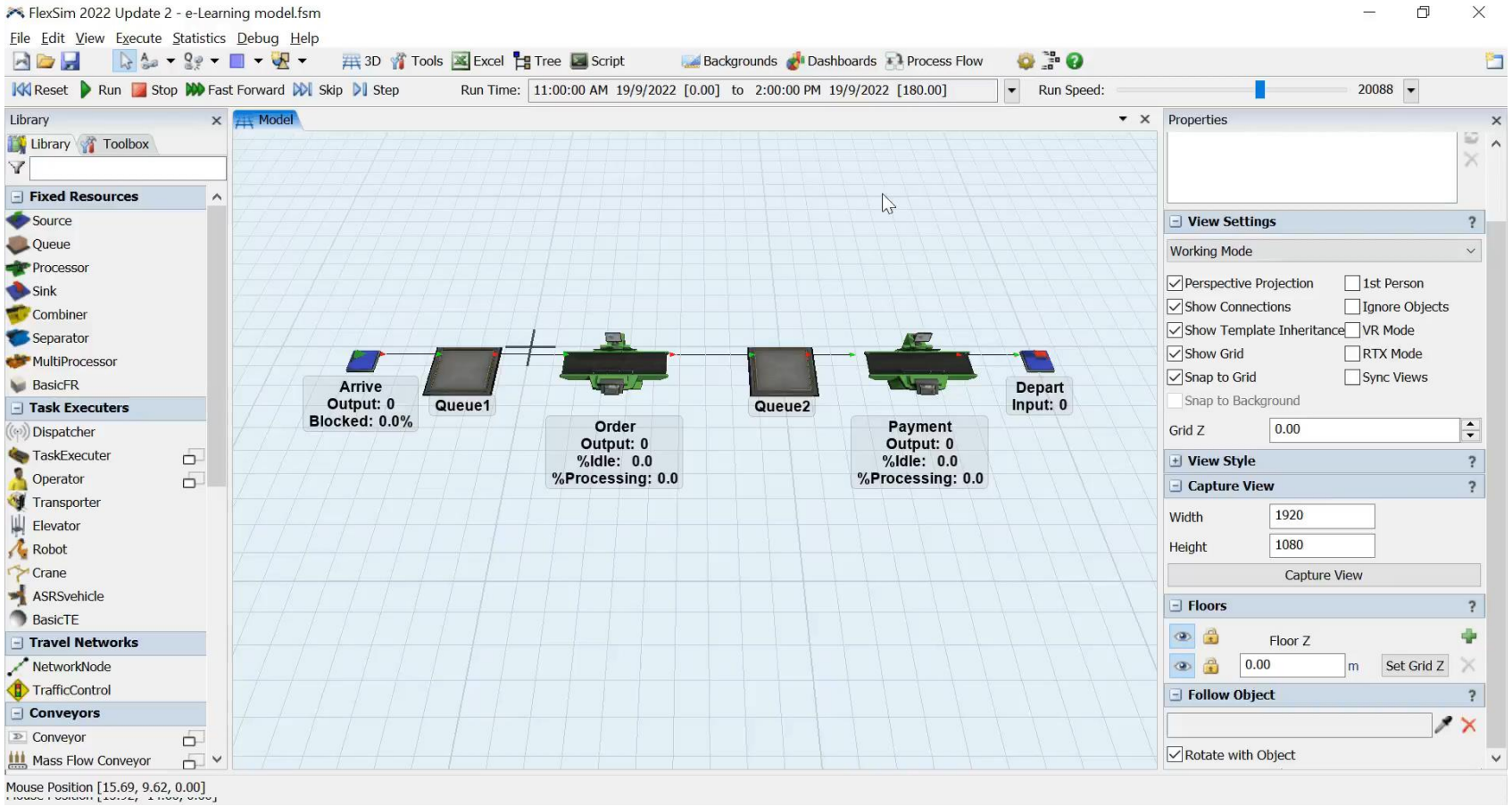
Building a Dashboard

Use Dashboard to Display Simulation Stats

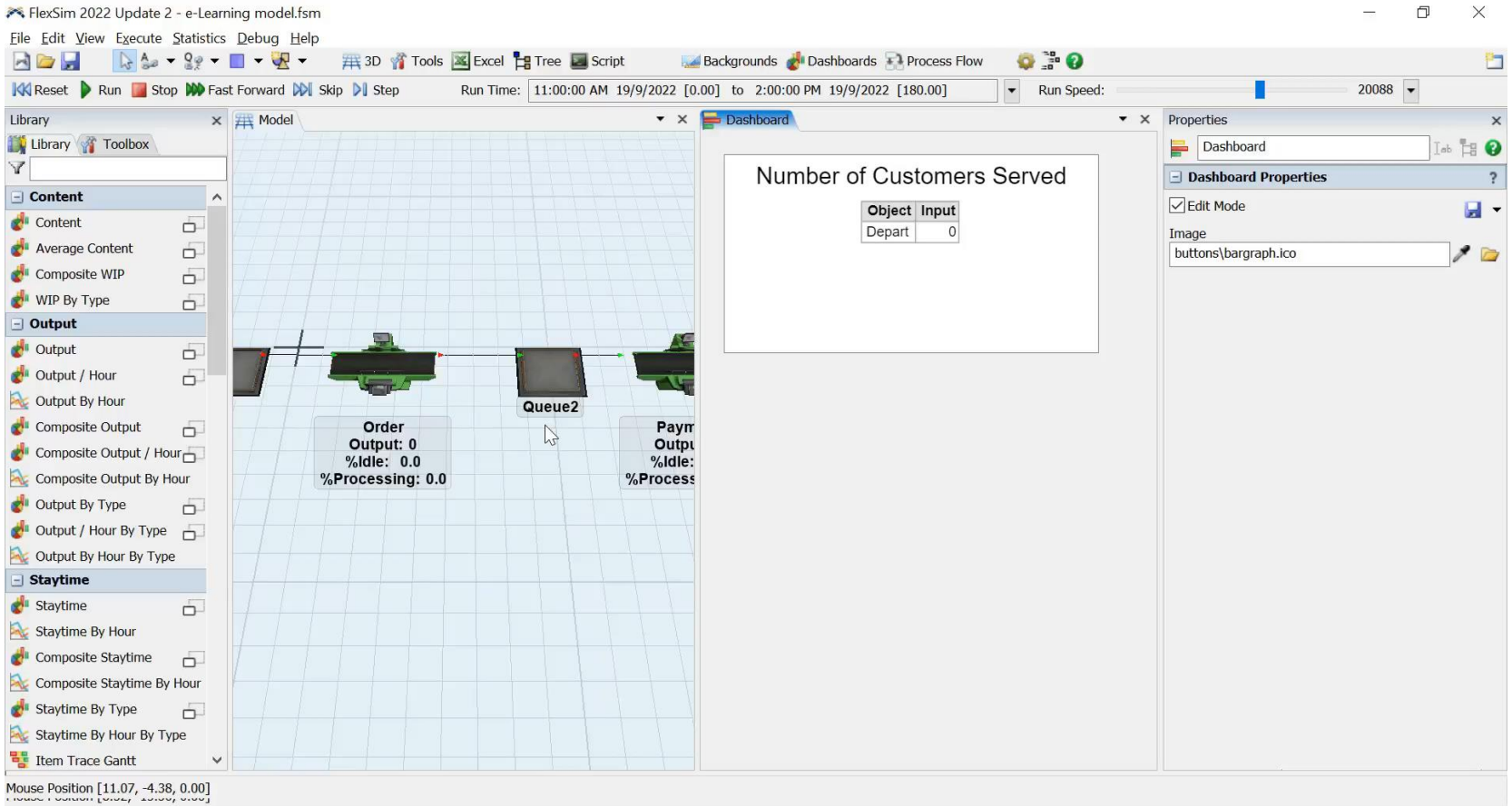
Add charts/ tables to show these results:

1. Number of customers served “DEPART” (‘Output’)
2. Average number in queue (‘Average Content’)
3. Average waiting time in queue (‘Staytime’)
4. Utilization of each processor (‘State’)
5. Time in system (‘Tracked Variable’)
6. Number in system (‘Composite WIP’)

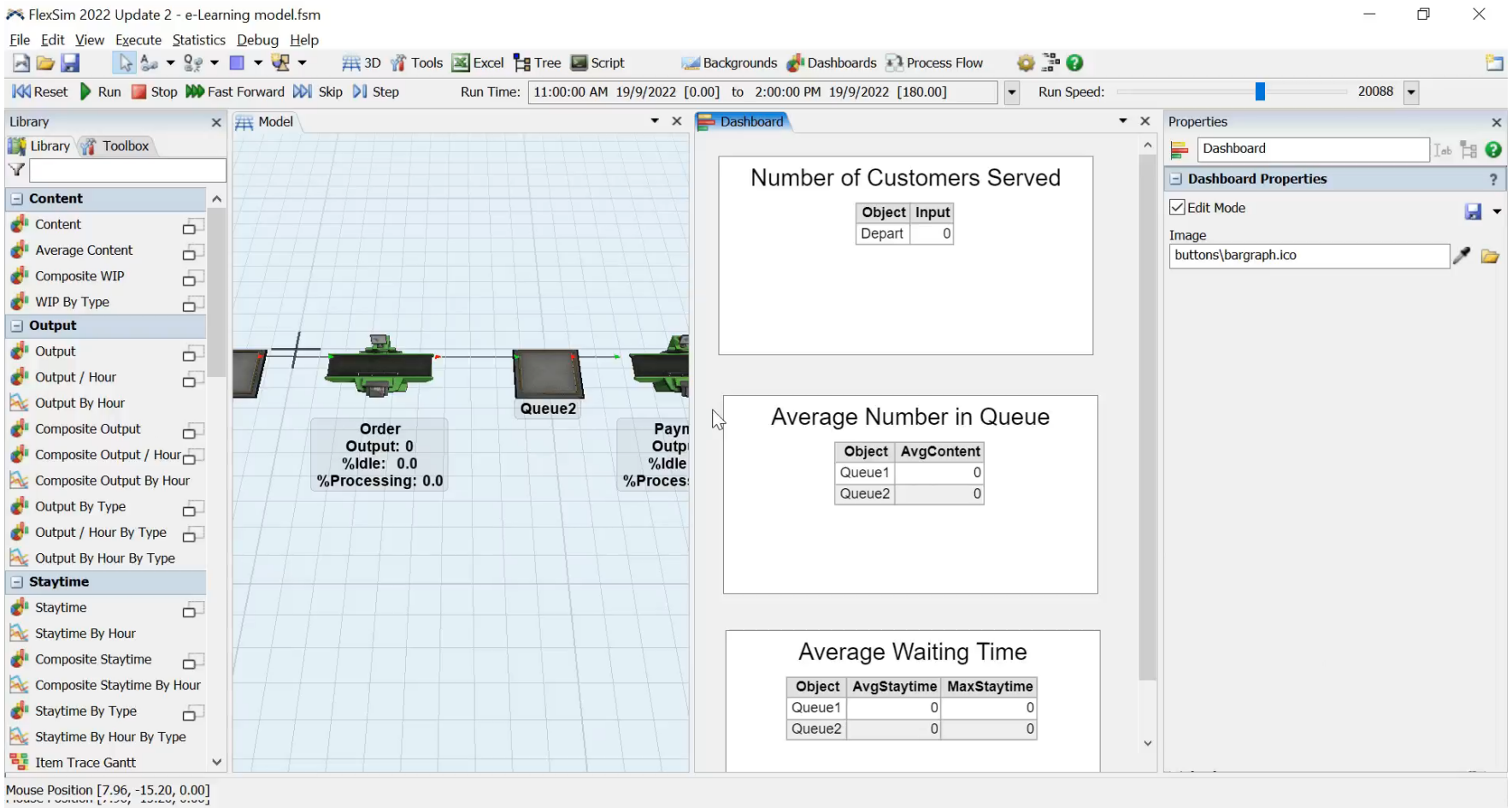
Adding a Dashboard & Number of Customers Served Table



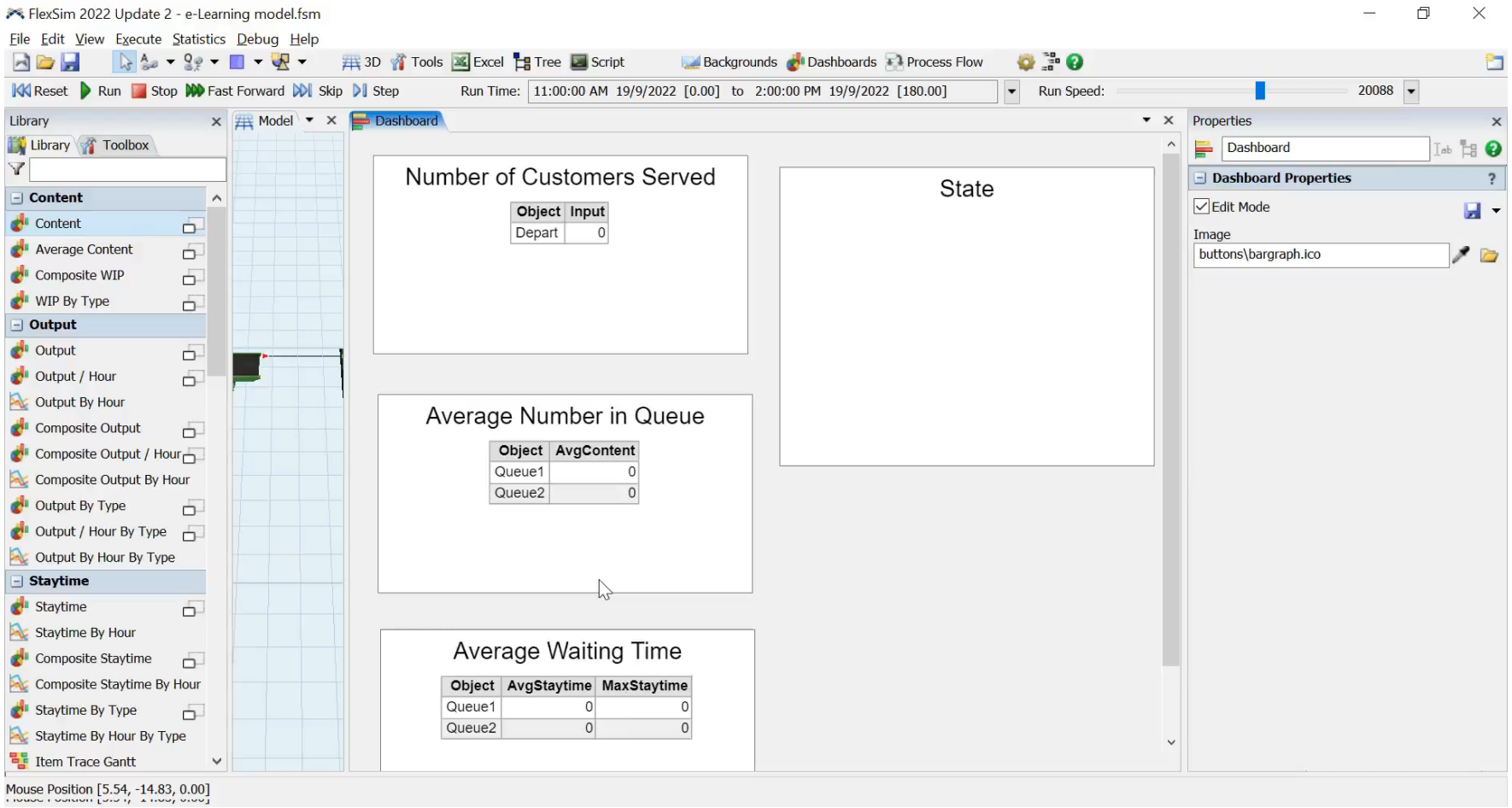
Average Queue Length & Waiting Time Tables



State (Utilization) Pie Chart



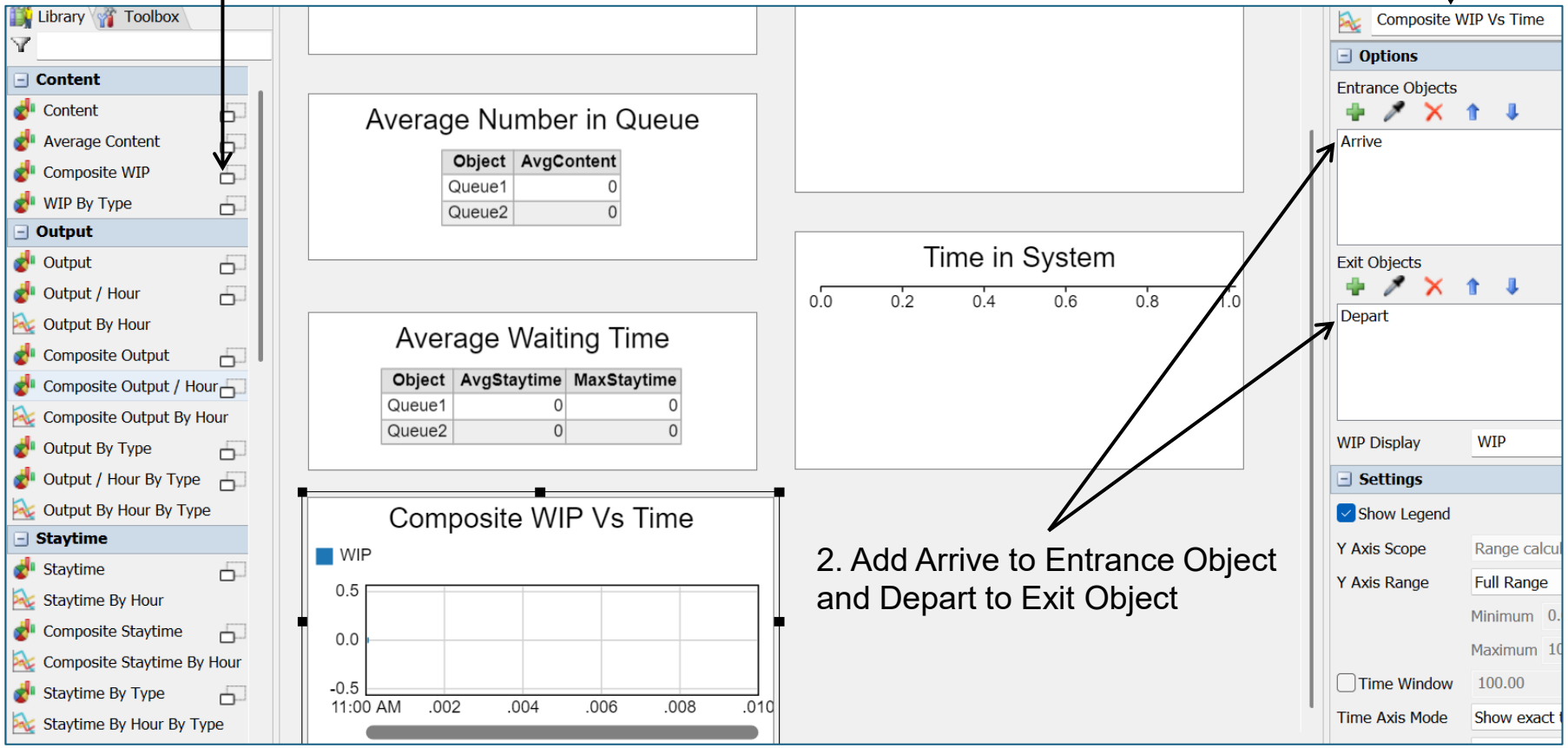
Time in System Chart



Number in System Chart

1. Add Composite WIP Line Chart

3. Rename as ‘Number in System Vs Time’



2. Add Arrive to Entrance Object and Depart to Exit Object

Simulation Dashboard Results

Number of Customers Served

Object	Input
Depart	107

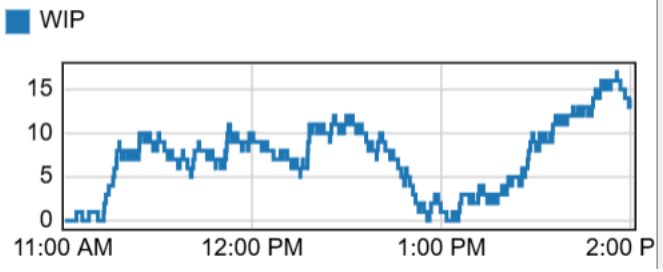
Average Number in Queue

Object	AvgContent
Queue1	5.21
Queue2	0.40

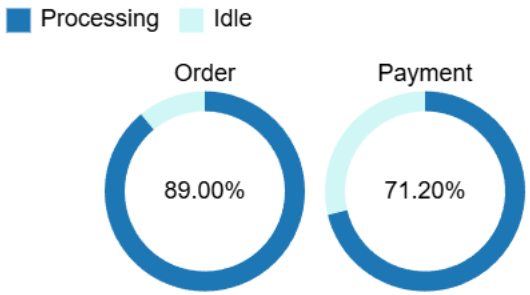
Average Waiting Time

Object	AvgStaytime	MaxStaytime
Queue1	7.48	20.23
Queue2	0.67	4.39

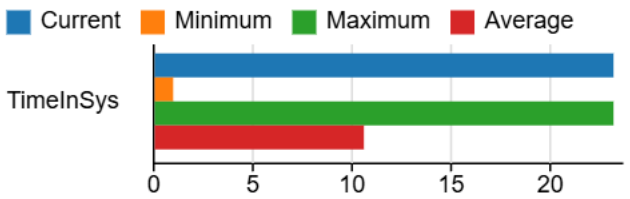
Number in System Vs Time



State

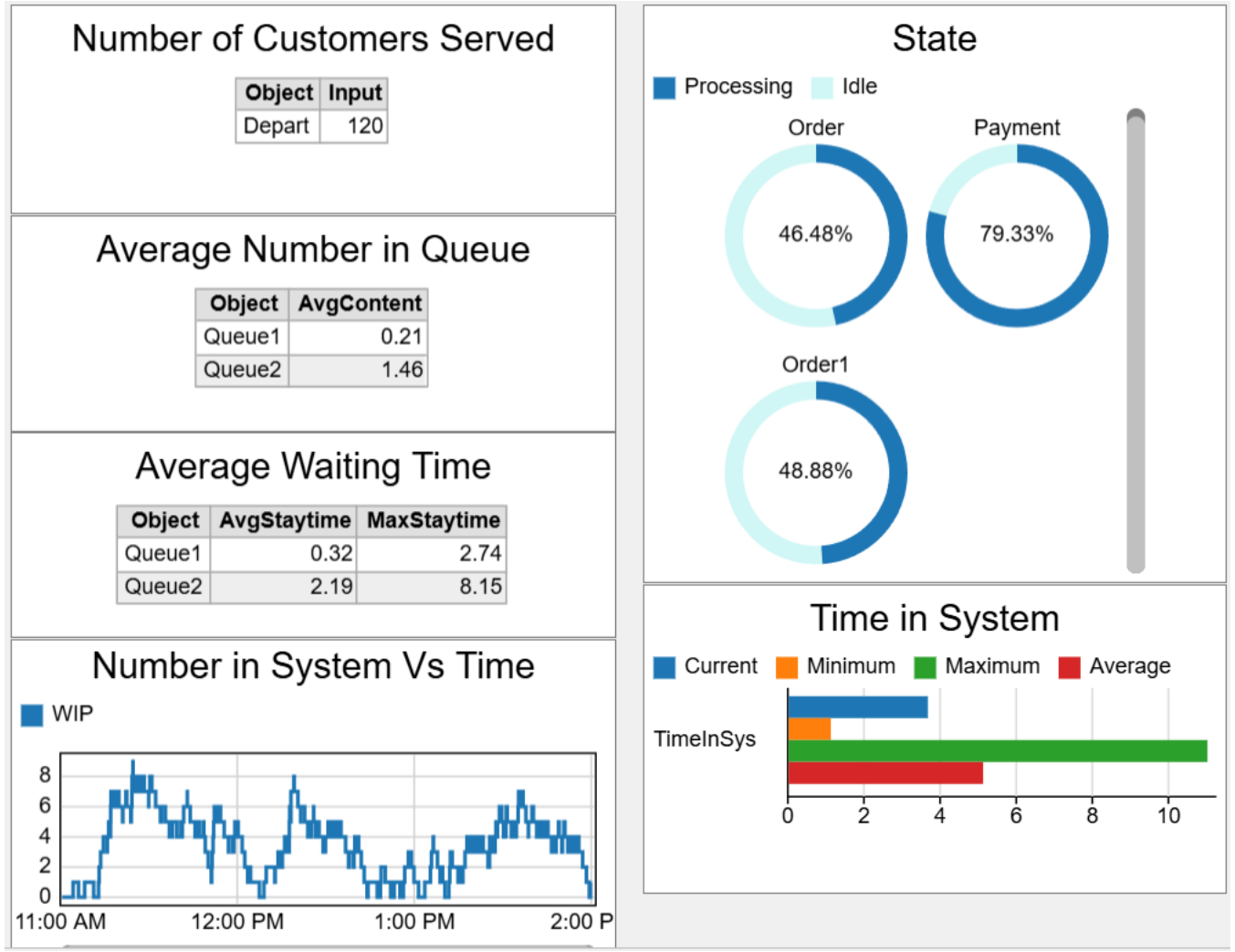


Time in System



Run the simulation till stop time. View the results in Dashboard.

Simulation Results with 2 Order Staff



Performance Measure	Change
Number of customers served	↑
Number of customers in system	↓
Time in system	↓

Which activity (order or payment) now has a longer queue, waiting time and utilization?

Experimenting with Conveyors

Types of Conveyors

- Used to convey flowitems from one location to another location.
- Time to convey depends on the conveyor's length divided by its speed.

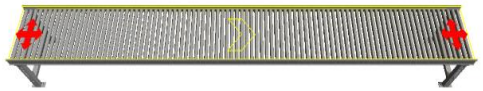
Name and Icon	Description	Appearance
Straight Conveyor	The **Straight Conveyor** can simulate conveyor belts or roller conveyors. As its name suggests, this conveyor has a straight shape.	
Curved Conveyor	The **Curved Conveyor** can also simulate conveyor belts or roller conveyors. Unlike the **Straight Conveyor**, this conveyor has a curved shape with varying radius settings.	

Creating Conveyors

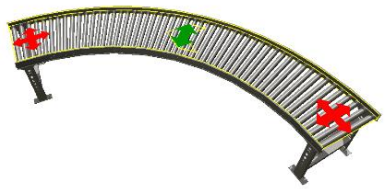
- The **Straight** and **Curved Conveyor** objects have a default size if you click and drag them into the model.

- You can set the initial lengths by clicking either the Straight Conveyor or the Curved Conveyor object under the Library. Your mouse pointer will change to a plus sign with either a Straight Conveyor icon or a Curved Conveyor icon next to it.

- You can also edit the length of an existing conveyor by clicking on the conveyor and dragging its cross arrows.



- Take note the direction of the conveyor flow represented by the arrow head in the middle.

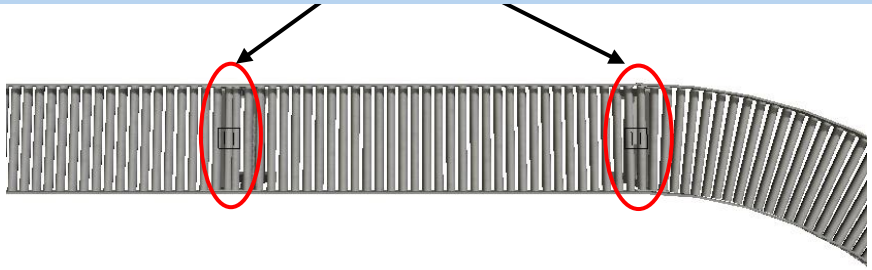


Connecting 2 Conveyors

To connect two conveyors together:

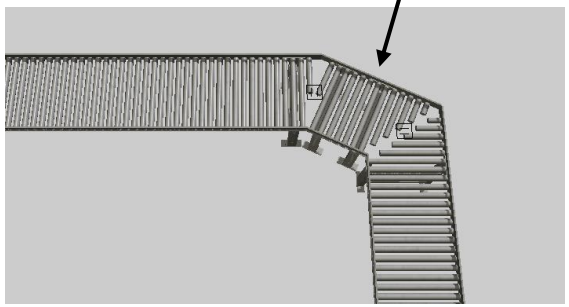
- Click on one conveyor and drag it close to the edge of the other conveyor. When the edges of the two conveyors get close enough, they will snap together.

Transfer across conveyors happen automatically

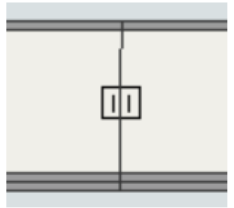
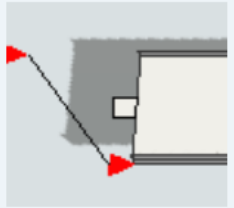
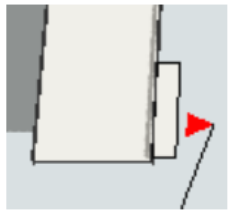


- To join 2 straight conveyors using a curved section, use ‘Join Conveyor’ and click one conveyor then the other to join them

Section formed by ‘Join Conveyor’



Connecting Conveyors and Objects

Name	Description	Appearance
Transfers	Transfers are connections from one conveyor to another. They can affect how an item is transferred between conveyors, especially when merging lanes together.	
Entry Transfers	Entry transfers are connections from a non-conveyor object (such as a source or other fixed resource) to a conveyor object. They can affect how an item is transferred to a conveyor.	
Exit Transfers	Exit transfers are connections from a conveyor to a non-conveyor object (such as a sink or other fixed resource). Exit transfers have the ability to send for a transport (a task executer) to pick up items and deliver them to another object. Task executers can pick up items from a particular point on the exit transfer or a range of possible points along an exit transfer.	

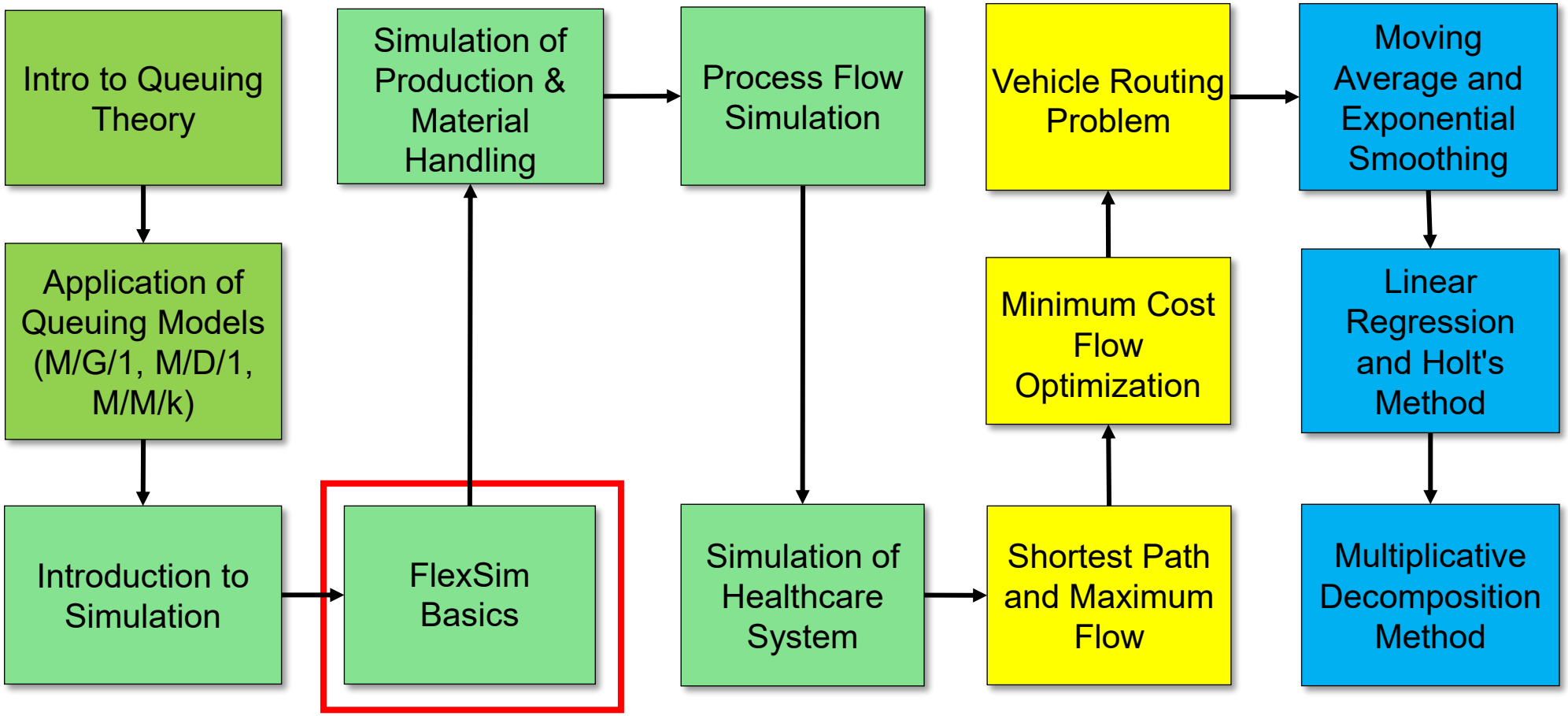
To connect conveyors to other objects, create a port connection using 'A' connect. This will create a new Entry Transfer or an Exit Transfer, an in-between object that is associated with a specific point or range on the conveyor.

An Exit Transfer is created when you connect a conveyor to another object.

An Entry Transfer is created when you connect an object to a conveyor.

Learning Outcomes

At the end of the lesson, students should be able to:

- Build a basic 3D simulation model
- Build simulation results dashboard
- Interpret simulation statistics

E211 Progression